

# Yukawaon model and unified description of quark and lepton mass matrices<sup>1</sup>

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## Abstract

In the so-called yukawaon model, where effective Yukawa coupling constants  $Y_f^{eff}$  ( $f = e, \nu, u, d$ ) are given by vacuum expectation values of gauge singlet scalars (yukawaons)  $Y_f$  with  $3 \times 3$  flavor components, it is tried to give a unified description of quark and lepton mass matrices. Especially, without assuming any discrete symmetry in the lepton sector, nearly tribimaximal mixing is derived by assumed a simple up-quark mass matrix form.

## 1 What is a yukawaon model?

First, let us give a short review of the so-called *yukawaon* model: We regard Yukawa coupling constants  $Y_f$  as effective coupling constants  $Y_f^{eff}$  in an effective theory, and we consider that  $Y_f^{eff}$  originate in vacuum expectation values (VEVs) of new gauge singlet scalars  $Y_f$ , i.e.

$$Y_f^{eff} = \frac{y_f}{\Lambda} \langle Y_f \rangle, \quad (1)$$

where  $\Lambda$  is a scale of an effective theory which is valid at  $\mu \leq \Lambda$ , and we assume  $\langle Y_f \rangle \sim \Lambda$ . We refer the fields  $Y_f$  as *yukawaons* [1] hereafter. Note that the effective coupling constants  $Y_f^{eff}$  evolve as those in the standard SUSY model below the scale  $\Lambda$ , since a flavor symmetry is completely broken at a high energy scale  $\mu \sim \Lambda$ .

In the present work, we assume an  $O(3)$  flavor symmetry. In order to distinguish each  $Y_f$  from others, we assume a  $U(1)_X$  symmetry (i.e. *sector charge*). (The  $SU(2)_L$  doublet fields  $q, \ell, H_u$  and  $H_d$  are assigned to sector charges  $Q_X = 0$ .) Then, we obtain VEV relations as follows: (i) We give an  $O(3)$  and  $U(1)_X$  invariant superpotential for yukawaons  $Y_f$ . (ii) We solve SUSY vacuum conditions  $\partial W / \partial Y_f = 0$ . (iii) Then, we obtain VEV relations among  $Y_f$ .

For example, in the seesaw-type neutrino mass matrix,  $M_\nu \propto \langle Y_\nu \rangle \langle Y_R \rangle^{-1} \langle Y_\nu \rangle^T$ , we obtain [2]

$$\langle Y_R \rangle \propto \langle Y_e \rangle \langle \Phi_u \rangle + \langle \Phi_u \rangle \langle Y_e \rangle \quad (2)$$

together with  $\langle Y_\nu \rangle \propto \langle Y_e \rangle$  and  $\langle Y_u \rangle \propto \langle \Phi_u \rangle \langle \Phi_u \rangle$ , i.e. a neutrino mass matrix is given by

$$\langle M_\nu \rangle_e \propto \langle Y_e \rangle_e \{ \langle Y_e \rangle_e \langle \Phi_u \rangle_e + \langle \Phi_u \rangle_e \langle Y_e \rangle_e \}^{-1} \langle Y_e \rangle_e, \quad (3)$$

where  $\langle \Phi_u \rangle_u \propto \text{diag}(\sqrt{m_u}, \sqrt{m_c}, \sqrt{m_t})$ , and  $\langle A \rangle_f$  denotes a form of a VEV matrix  $\langle A \rangle$  in the diagonal basis of  $\langle Y_f \rangle$  (we refer it as  $f$  basis). We can obtain a form  $\langle \Phi_u \rangle_d = V(\delta)^T \langle \Phi_u \rangle_u V(\delta)$

<sup>1</sup> To appear in the Proceedings of Lepton-Photon 2009.

from the definition of the CKM matrix  $V(\delta)$ , but we do not know an explicit form of  $\langle \Phi_u \rangle_e$ . Therefore, in a previous work [2], the author put an ansatz,  $\langle \Phi_u \rangle_e = V(\pi)^T \langle \Phi_u \rangle_u V(\pi)$  by supposing  $\langle \Phi_u \rangle_e \simeq \langle \Phi_u \rangle_d$ , and he obtained excellent predictions of the neutrino oscillation parameters without assuming any discrete symmetry. However, there is no theoretical ground for the ansatz for the form  $\langle \Phi_u \rangle_e$ .

The purpose of the present work is to investigate a quark mass matrix model in order to predict neutrino mixing parameters on the basis of a yukawaon model (2), without such the ad hoc ansatz, because if we give a quark mass matrix model where mass matrices  $(M_u, M_d)$  are given on the  $e$  basis, then, we can obtain the form  $\langle \Phi_u \rangle_e$  by using a transformation  $\langle \Phi_u \rangle_e = U_u \langle \Phi_u \rangle_u U_u^T$ , where  $U_u$  is defined by  $U_u^T M_u U_u = M_u^{diag}$ .

## 2 Yukawaons in the quark sector

We assume a superpotential in the quark sector [3]:

$$W_q = \mu_u [Y_u \Theta_u] + \lambda_u [\Phi_u \Phi_u \Theta_u] + \mu_u^X [\Phi_u \Theta_u^X] + \mu_d^X [Y_d \Theta_d^X] + \sum_{q=u,d} \frac{\xi_q}{\Lambda} [\Phi_e (\Phi_X + a_q E) \Phi_e \Theta_q^X]. \quad (4)$$

Here and hereafter, for convenience, we denotes  $\text{Tr}[\dots]$  as  $[\dots]$  simply. From SUSY vacuum conditions  $\partial W / \partial \Theta_u = 0$ ,  $\partial W / \partial \Theta_u^X = 0$  and  $\partial W / \partial \Theta_d^X = 0$ , we obtain  $\langle Y_u \rangle \propto \langle \Phi_u \rangle \langle \Phi_u \rangle$ ,

$$M_u^{1/2} \propto \langle \Phi_u \rangle_e \propto \langle \Phi_e \rangle_e (\langle \Phi_X \rangle_e + a_u \langle E \rangle_e) \langle \Phi_e \rangle_e, \quad (5)$$

$$M_d \propto \langle Y_d \rangle_e \propto \langle \Phi_e \rangle_e (\langle \Phi_X \rangle_e + a_d \langle E \rangle_e) \langle \Phi_e \rangle_e, \quad (6)$$

respectively. Here,  $\langle \Phi_X \rangle_e$  and  $\langle E \rangle_e$  are given by

$$\langle \Phi_X \rangle_e \propto X \equiv \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \langle E \rangle_e \propto \mathbf{1} \equiv \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (7)$$

(Note that the VEV form  $\langle \Phi_X \rangle_e$  breaks the  $O(3)$  flavor symmetry into  $S_3$ .) Therefore, we obtain quark mass matrices

$$M_u^{1/2} \propto M_e^{1/2} (X + a_u \mathbf{1}) M_e^{1/2}, \quad M_d \propto M_e^{1/2} (X + a_d e^{i\alpha_d} \mathbf{1}) M_e^{1/2}, \quad (8)$$

on the  $e$  basis. Note that we have assumed that the  $O(3)$  relations are valid only on the  $e$  and  $u$  bases, so that  $\langle Y_e \rangle$  and  $\langle Y_u \rangle$  must be real.

A case  $a_u \simeq -0.56$  can give a reasonable up-quark mass ratios  $\sqrt{m_{u1}/m_{u2}} = 0.043$  and  $\sqrt{m_{u2}/m_{u3}} = 0.057$ , which are in favor of the observed values [4]  $\sqrt{m_u/m_c} = 0.045_{-0.010}^{+0.013}$ , and  $\sqrt{m_c/m_t} = 0.060 \pm 0.005$  at  $\mu = M_Z$ .

| Sector      | Parameters          | Predictions                                                                              |                                   |            |
|-------------|---------------------|------------------------------------------------------------------------------------------|-----------------------------------|------------|
|             |                     | $\sin^2 \theta_{atm}$                                                                    | $\tan^2 \theta_{solar}$           | $ U_{13} $ |
| $M_\nu$     | $\xi = +0.0005$     | 0.982                                                                                    | 0.449                             | 0.012      |
|             | $\xi = -0.0012$     | 0.990                                                                                    | 0.441                             | 0.014      |
| $M_u^{1/2}$ | $a_u = -0.56$       | $\sqrt{\frac{m_u}{m_c}} = 0.0425$                                                        | $\sqrt{\frac{m_c}{m_t}} = 0.0570$ |            |
|             | two parameters      | 5 observables: fitted excellently                                                        |                                   |            |
| $M_d$       | $a_d e^{i\alpha_d}$ | $\sqrt{\frac{m_d}{m_s}}, \sqrt{\frac{m_s}{m_b}},  V_{us} ,  V_{cb} ,  V_{ub} ,  V_{td} $ |                                   |            |
|             | two parameters      | 6 observables: not always excellent                                                      |                                   |            |

Table 1: Summary of the present model.

### 3 Yukawaons in the neutrino

However, the up-quark mass matrix (5) failed to give reasonable neutrino oscillation parameter values although it can give reasonable up-quark mass ratios. Therefore, we will slightly modify the model (2) in the neutrino sector.

Note that the sign of the eigenvalues of  $M_u^{1/2}$  given by Eq.(8) is  $(+, -, +)$  for the case  $a_u \simeq -0.56$ . If we assume that the eigenvalues of  $\langle \Phi_u \rangle_u$  must be positive, so that  $\langle \Phi_u \rangle_u$  in Eq.(2) is replaced as  $\langle \Phi_u \rangle_u \rightarrow \langle \Phi_u \rangle_u \cdot \text{diag}(+1, -1, +1)$ , then, we can obtain successful results except for  $\tan^2 \theta_{solar}$ , i.e. predictions  $\sin^2 2\theta_{atm} = 0.984$  and  $|U_{13}| = 0.0128$  and an unfavorable prediction  $\tan^2 \theta_{solar} = 0.7033$ .

When we introduce a new field  $P_u$  with a VEV  $\langle P_u \rangle_u \propto \text{diag}(+1, -1, +1)$ , we must consider an existence of  $P_u Y_e \Phi_u + \Phi_u Y_e P_u$  in addition to  $Y_e P_u \Phi_u + \Phi_u P_u Y_e$ , because they have the same  $U(1)_X$  charges. Therefore, we modify Eq.(2) into

$$W_R = \mu_R [Y_R \Theta_R] + \frac{\lambda_R}{\Lambda} \{ [(Y_e P_u \Phi_u + \Phi_u P_u Y_e) \Theta_R] + \xi [(P_u Y_e \Phi_u + \Phi_u Y_e P_u) \Theta_R] \}, \quad (9)$$

which leads to VEV relation  $Y_R \propto Y_e P_u \Phi_u + \Phi_u P_u Y_e + \xi (P_u Y_e \Phi_u + \Phi_u Y_e P_u)$ . The results at  $a_u \simeq -0.56$  are excellently in favor of the observed neutrino oscillation parameters by taking a small value of  $|\xi|$  (see Table 1):

Also, we can calculate the down-quark sector. We have two parameters  $(a_d, \alpha_d)$  in the down-quark sector given in Eq.(8). (See Table 2 in Ref.[3]). The results are roughly reasonable, although  $|V_{13}|$  and  $|V_{3i}|$  are somewhat larger than the observed values. Those discrepancies will be improved in future version of the model.

### 4 Summary

In conclusion, for the purpose of deriving the observed nearly tribimaximal neutrino mixing, a possible yukawaon model in the quark sector is investigated. Five observable quantities (2 up-quark mass ratios and 3 neutrino mixing parameters) are excellently fitted by two parameters. Also, the CKM mixing parameters and down-quark mass ratios are given under additional 2 parameters. The results are summarized in Table 1.

It is worthwhile to notice that the observed tribimaximal mixing in the neutrino sector is substantially obtained from the up-quark mass matrix structure (8). Although the model for down-quark sector still need an improvement, the present approach will provide a new view to a unified description of the masses and mixings.

## References

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